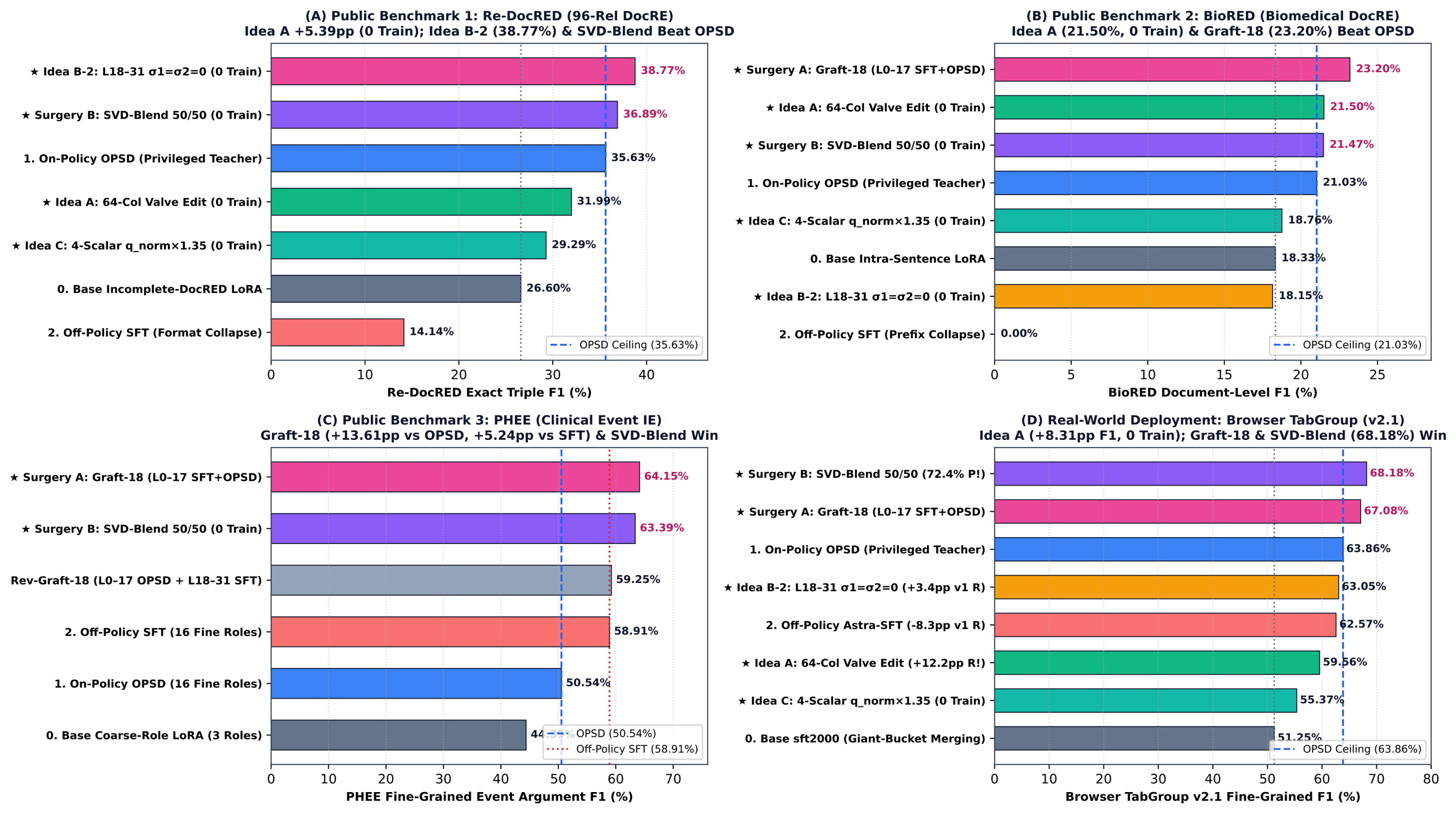
(A) Public Benchmark 1: Re-DocRED (96-Rel DocRE)
Idea A +5.39pp (0 Train); Idea B-2 (38.77%) & SVD-Blend Beat OPSD
★ Idea B-2: L18-31 σ1=σ2=0 (0 Train)
38.77%
★ Surgery B: SVD-Blend 50/50 (0 Train)
36.89%
1. On-Policy OPSD (Privileged Teacher)
35.63%
★ Idea A: 64-Col Valve Edit (0 Train)
31.99%
★ Idea C: 4-Scalar q_norm×1.35 (0 Train)
29.29%
0. Base Incomplete-DocRED LoRA
26.60%
2. Off-Policy SFT (Format Collapse)
14.14%
OPSD Ceiling (35.63%)
0
10
20
30
40
Re-DocRED Exact Triple F1 (%)
(B) Public Benchmark 2: BioRED (Biomedical DocRE)
Idea A (21.50%, 0 Train) & Graft-18 (23.20%) Beat OPSD
★ Surgery A: Graft-18 (L0-17 SFT+OPSD)
23.20%
★ Idea A: 64-Col Valve Edit (0 Train)
21.50%
★ Surgery B: SVD-Blend 50/50 (0 Train)
21.47%
1. On-Policy OPSD (Privileged Teacher)
21.03%
★ Idea C: 4-Scalar q_norm×1.35 (0 Train)
18.76%
0. Base Intra-Sentence LoRA
18.33%
★ Idea B-2: L18-31 σ1=σ2=0 (0 Train)
18.15%
2. Off-Policy SFT (Prefix Collapse)
0.00%
OPSD Ceiling (21.03%)
0
5
10
15
20
25
BioRED Document-Level F1 (%)
(C) Public Benchmark 3: PHEE (Clinical Event IE)
Graft-18 (+13.61pp vs OPSD, +5.24pp vs SFT) & SVD-Blend Win
★ Surgery A: Graft-18 (L0-17 SFT+OPSD)
64.15%
★ Surgery B: SVD-Blend 50/50 (0 Train)
63.39%
Rev-Graft-18 (L0-17 OPSD + L18-31 SFT)
59.25%
2. Off-Policy SFT (16 Fine Roles)
58.91%
1. On-Policy OPSD (16 Fine Roles)
50.54%
0. Base Coarse-Role LoRA (3 Roles)
OPSD (50.54%)
Off-Policy SFT (58.91%)
0
10
20
30
40
50
60
70
PHEE Fine-Grained Event Argument F1 (%)
(D) Real-World Deployment: Browser TabGroup (v2.1)
Idea A (+8.31pp F1, 0 Train); Graft-18 & SVD-Blend (68.18%) Win
★ Surgery B: SVD-Blend 50/50 (72.4% P!)
68.18%
★ Surgery A: Graft-18 (L0-17 SFT+OPSD)
67.08%
1. On-Policy OPSD (Privileged Teacher)
63.86%
★ Idea B-2: L18-31 σ1=σ2=0 (+3.4pp v1 R)
63.05%
2. Off-Policy Astra-SFT (-8.3pp v1 R)
62.57%
★ Idea A: 64-Col Valve Edit (+12.2pp R!)
59.56%
★ Idea C: 4-Scalar q_norm×1.35 (0 Train)
55.37%
0. Base sft2000 (Giant-Bucket Merging)
51.25%
OPSD Ceiling (63.86%)
0
10
20
30
40
50
60
70
80
Browser TabGroup v2.1 Fine-Grained F1 (%)